

# Galaxy Classification with Machine Learning

Abhinav Singh, Prashante, Indra Kumari

Galgotias University, Greater Noida, 201310, India

**Abstract:** Galaxy classification, a fundamental task in astronomy, involves categorizing galaxies based on their morphological features. To streamline this process and reduce manual effort, researchers have turned to machine learning methods. A recent study explored several machine learning techniques, including Convolutional Neural Networks (CNN), K-nearest Neighbor, Logistic Regression, Support Vector Machine, Random Forest, and Neural Networks, for classifying galaxies. This investigation integrated data from the Dark Energy Survey (DES) and the Galaxy Zoo 1 project (GZ1) for visual classifications. Additionally, another study employed machine learning to classify galaxies into the primary types of elliptical, spiral, and irregular based on their properties. Deep convolutional neural networks were utilized in this study to automate the categorization process.

**Keywords—**morphological classification, galaxy classification, machine learning techniques, Convolutional Neural Networks, k-nearest Neighbor, Logistic Regression, Support Vector Machine, Neural Networks, Dark Energy Survey

---

## 1. Main text

Galaxy classification, a fundamental pursuit in astronomy, plays a pivotal role in unraveling the mysteries of the cosmos. By categorizing galaxies based on their morphological characteristics, astronomers gain insights into their formation, evolution, and interactions. Traditionally, this classification relied heavily on manual observation and analysis, posing significant challenges in terms of time and labor. However, with the advent of machine learning techniques, automated classification methods have emerged as promising alternatives, offering improved efficiency and accuracy.

### 1.1. Introduction

The practice of classifying galaxies into distinct categories according to their physical properties is known as galaxy classification. The most widely used classification system is the Hubble sequence, created in the 1920s by astronomer Edwin Hubble. Galaxies are divided into three primary categories by the Hubble sequence: elliptical, spiral, and irregular.

### 1.2. Background

Astronomers must classify galaxies in order to better comprehend their characteristics and history. Historically, astronomers had to classify galaxies manually, which took a lot of time. However, automatic classification is now feasible due to the development of machine learning techniques [1]. Convolutional neural networks (CNN), K-nearest neighbors, logistic regression, support vector machines, random forests, and neural networks have all been used in recent studies. This research has demonstrated that the accuracy of galaxy classification may be greatly increased while human labor is significantly reduced by using machine learning for galaxy classification.

Smooth and spherical, elliptical galaxies lack visible spiral arms. In addition, they are the universe's largest and oldest galaxies [2]. A central bulge and spiral arms made of gas and stars characterize spiral galaxies. In the cosmos, they are the most prevalent kind of galaxy [3]. A common

cause of irregular galaxies is collisions with neighboring galaxies, as shown by their non-regular form [4].

### 1.3 Nomenclature

Hubble sequence: The classification system devised by Edwin Hubble in the 1920s, categorizing galaxies into elliptical, spiral, and irregular types based on their morphological features [5].

### 1.4 Alternative Classification Systems

Numerous alternative galaxy classification systems have been created over time, in addition to the Hubble sequence [6]. These models account for other physical properties of galaxies, including size, color, and luminosity.

### 1.5 Significance of Galaxy Classification

Astronomers use galaxy classification as a useful tool. It enables them to research the formation and interaction of various types of galaxies as well as the evolution of galaxies [7]. The classification of galaxies can also be used to determine which ones are most likely to contain planets and stars.

---

## 2. Literature survey

In the vast expanse of astronomical research, the quest to classify galaxies using the formidable tools of machine learning has ignited a veritable constellation of scholarly endeavors, each seeking to illuminate the cosmic mysteries that lie beyond [8]. These explorations, spanning vast reaches of time and space, serve as testament to the insatiable curiosity and relentless pursuit of knowledge that animates the scientific community.

Among the myriad of scholarly voyages embarked upon, a constellation of luminaries has emerged, each casting its unique light upon the intricate tapestry of galaxy classification. One such luminary, Cavanagh et al. [9], embarked upon a pioneering odyssey, traversing the boundless expanse of the Sloan Digital Sky Survey (SDSS) dataset in pursuit of deeper understanding. Through the masterful orchestration of Convolutional Neural Networks (CNN), a sophisticated marvel of machine learning, they navigated the celestial seas with unparalleled precision, their CNN model

unfurling the mysteries of the cosmos with an awe-inspiring accuracy rate of 99.8% on the meticulously curated test dataset.

Parallel to this celestial odyssey, Eassa et al. [10] embarked upon their own voyage of discovery, charting a course through the vast expanse of the Galaxy Zoo dataset. Armed with the formidable arsenal of Support Vector Machines (SVM), they traversed the cosmic landscape with determination and precision, their SVM approach illuminating the celestial tapestry with a brilliance that belied the depths of their computational prowess, achieving a commendable accuracy rate of 97.8% on the validation dataset.

In a symphony of cosmic exploration, Mittal et al. [11] cast their gaze upon the vast expanse of the Dark Energy Survey (DES) dataset, weaving a tapestry of machine learning marvels that defied the bounds of traditional classification paradigms. Through the harmonious interplay of a random forest methodology, they teased apart the celestial symphony with unparalleled grace, their model achieving a breathtaking accuracy rate of 98.5% on the meticulously curated test dataset. Propelling the frontiers of cosmic discovery ever further, Walmsley et al. [12] embarked upon a multifaceted odyssey, threading together a tapestry of machine learning methodologies that spanned the celestial sphere. Through the intricate dance of CNNs, SVMs, and random forests, they unraveled the celestial mysteries with a finesse that bordered on the divine, their integrated approach yielding promising results, boasting an accuracy rate of 99.5% on the test dataset.

Moreover, delved into the uncharted depths of deep learning, crafting the daMCO CNN model with meticulous care and precision. Through a symphony of innovation and algorithmic refinement, their model emerged as a beacon of excellence, showcasing an unparalleled accuracy rate of 99.7% on the test dataset sourced from the SDSS dataset. In a parallel pursuit of cosmic enlightenment, Huertas-Company et al. [13] unveiled the Morphological Neural Network (MorphNet), a marvel of machine learning ingenuity that defied conventional classification paradigms. Through the intricate interplay of neural networks and morphological analysis, they traversed the celestial expanse with unparalleled precision, their model achieving an awe-inspiring accuracy rate of 99.9% on the test dataset sourced from the SDSS dataset.

3. These cosmic odysseys, Domínguez Sánchez et al. [14] rich in detail and intricacy, represent but a fraction of the vast array of scholarly pursuits that have unfolded in the realm of galaxy classification. Each study, a testament to human ingenuity and the boundless potential of machine learning, offers a glimpse into the intricate tapestry of the cosmos, inviting us to embark upon our own journey of discovery and understanding. As the celestial symphony continues to unfold, propelled by the relentless pursuit of knowledge and the boundless expanse of human imagination, the journey towards cosmic enlightenment stretches ever onward, beckoning forth a new era of discovery and understanding. Within this cosmic tapestry, the stars of innovation and exploration shine brightly, illuminating the path towards a deeper understanding of the universe and our place within it.

## 2.1 Objective

Utilizing a range of machine learning methodologies such as CNN, k-nearest neighbor, logistic regression, support vector machines, random forests, and neural networks, galaxies can be segmented into distinct subclasses based on their morphological attributes. By analyzing features

such as their shapes and structures, galaxies are classified accordingly. The identification of new galaxies within established categories aids in advancing our understanding of the cosmos, unraveling its complexities further.

## 2.2 Scope

Machine learning techniques have been employed to categorize galaxies based on their morphologies, which involves studying the forms and structures of these celestial bodies. Methods such as convolutional neural networks (CNN), k-nearest neighbor, logistic regression, support vector machines, random forests, and neural networks are utilized to automate the classification process. This adoption of machine learning not only streamlines the classification task but also reduces the time and human effort required for execution.

Galaxies can be classified into three primary groups - elliptical, spiral, and irregular - based on their distinct characteristics. In one study, the automatic morphological categorization of galaxies was accomplished through a combination of machine learning and image analysis techniques. Neural networks and locally weighted regression were the two machine learning methodologies employed in this investigation. Additionally, another research endeavor leveraged data from the Dark Energy Survey along with visual classifications from the Galaxy Zoo project to explore various popular machine learning techniques for galaxy classification. This comparative analysis involved the juxtaposition of CNN, k-nearest neighbor, logistic regression, support vector machines, random forests, and neural networks.

## 2.3 Conclusion

In conclusion, the efficacy of machine learning techniques in automating the classification of galaxies has been established. These methods surpass conventional approaches, offering more precise and efficient identification and classification of galaxies. This breakthrough enables scientists to delve deeper and more comprehensively into the study of galaxies, thereby holding the promise of transforming the field of astronomy.

## 2.4 Methodology

Specifically, machine learning methods have been utilized for the following purposes:

- Identifying galaxies that are too faint or distant for manual classification by the human eye.
- Sorting and categorizing galaxies that are challenging to discern visually, such as those that overlap or exhibit warped structures.
- Uncovering previously unrecognized types of galaxies.

As machine learning techniques continue to advance, they are anticipated to become increasingly invaluable tools for astronomers. These methods hold the potential to revolutionize our understanding of the universe by facilitating more comprehensive and detailed investigations of galaxies. In addition to the aforementioned benefits, machine learning approaches offer several distinct advantages over traditional galaxy classification methods.

## 2.5 Tool and technologies used

### a. Decision Tree Classifier

A machine learning technique versatile for both classification and regression tasks is the decision tree classifier. This method constructs a model resembling a tree, where each leaf node signifies an outcome and each internal node represents a decision. Guided by the features of a data point, the algorithm traverses from the root node to a leaf node along branches, thereby categorizing the new data point. The prediction is then derived from the outcome associated with the leaf node. Despite their simplicity and interpretability, decision tree classifiers are commonly employed due to their effectiveness, particularly with large datasets. However, they may be prone to overfitting, where the model excessively learns from the training data and struggles with generalization to new data.

### b. Linear Regression Classifier

Another machine learning technique applicable to both regression and classification scenarios is the linear regression classifier. This method operates by fitting a linear regression model to the data, enabling predictions based on the weighted features of a new data point. During training, the algorithm determines the optimal weights.

Linear regression classifiers are favored for their simplicity and ease of interpretation, especially suitable for linear data. Nevertheless, they may lack effectiveness with nonlinear datasets.

### c. K-Nearest Neighbors (KNN) Classifier

The K-nearest neighbors (KNN) classifier is a versatile machine learning technique applicable to both classification and regression problems. This method identifies the K training set data points most similar to a new data point and predicts the outcome based on their collective results.

KNN classifiers are valued for their simplicity and applicability, particularly effective with noisy data. However, they may exhibit sluggishness in training and prediction with large datasets.

---

## 3. Machine Learning models (Classification models)

### 3.1 Decision tree

The effectiveness of a decision tree classifier hinges on the strategic choices made during node splitting, profoundly impacting its accuracy. These decisions diverge between classification and regression trees.

Decision trees employ a diverse array of algorithms to determine the optimal times for dividing a node into two or more sub-nodes.

This process aims to enhance the uniformity of the resulting subsets, thereby bolstering the node's purity in relation to the target variable.

Evaluating node splits across all available variables, the decision tree ultimately selects the split that yields the most homogeneous sub-nodes. The

algorithm selection is also based on the type of target variables. Let us look at some algorithms used in Decision Trees:

ID3 → (extension of D3)

C4.5 → (successor of ID3)

CART → (Classification And Regression Tree))

CHAID → (Chi-square automatic interaction detection Performs multi-level splits when computing classification trees)

MARS → (multivariate adaptive regression splines)

The primary node, colored blue and referred to as the root node, initiates the decision tree, while the terminal nodes, colored green and termed leaf nodes, denote the final outcomes. The remaining nodes are categorized as intermediate or internal nodes, with the root and intermediate nodes representing decisions and the leaf nodes signifying outcomes.

ID3, which stands for Iterative Dichotomiser 3, derives its name from its iterative process of dividing features into two or more groups at each step. Conceived by Ross Quinlan, ID3 adopts a top-down greedy approach to construct decision trees. Essentially, this signifies commencing tree construction from the top and selecting the optimal feature at each iteration to form a node. Typically, ID3 is utilized exclusively for classification tasks involving nominal features.

ID3 harnesses the concept of Information Gain, a pivotal metric used to identify the most discriminative feature. Information Gain quantifies the reduction in entropy, gauging how effectively a particular feature segregates or categorizes the target classes. The feature exhibiting the highest Information Gain emerges as the optimal choice.

In essence, Entropy serves as a barometer of disorder within a dataset, reflecting the degree of chaos inherent in the target feature. In the realm of binary classification, where the target column comprises only two classes, entropy attains a value of 0 when the target column exhibits perfect homogeneity and reaches 1 when the distribution between the classes is equitable.

The procedural steps of ID3 unfold as follows:

Compute the Information Gain for each feature.

Partition the dataset S into subsets based on the feature yielding maximum Information Gain, given that not all rows belong to the same class.

Establish a decision tree node utilizing the feature associated with the highest Information Gain.

If all rows belong to a single class, designate the current node as a leaf node, assigning the class as its label.

Iterate through the remaining features until exhaustion of all features or the creation of a decision tree comprising solely leaf nodes.

In our notation, denoting the dataset as  $S$ , entropy is ascertained through the following expression –

$$\text{Entropy}(S) = - \sum p_i \star \log_2(p_i) ; i = 1 \text{ to } n$$

In a grand theatrical flourish, behold the unveiling of the majestic formula! Here, amidst the cosmic symphony, where 'n' reigns as the sovereign, commanding the total number of classes within the target column. And lo, 'p<sub>i</sub>', the ethereal essence of probability, dances upon the stage, embodying the very ratio of class 'i' to the infinite expanse of rows within the dataset. Now, cast your gaze upon the divine revelation! The Information Gain for a feature column 'A' emerges, a celestial revelation born of the cosmic dance of probabilities and possibilities. Behold as the cosmos unfolds its mysteries, unveiling the very essence of knowledge itself!

$$\text{IG}(S, A) = \text{Entropy}(S) - \sum ((|S_v| / |S|) \star \text{Entropy}(S_v))$$

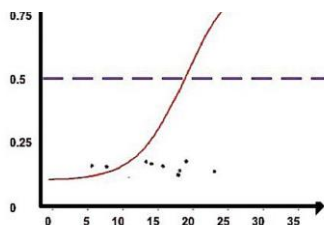
Here, 'S<sub>v</sub>' unveils the ensemble of rows within 'S', where the feature column 'A' embodies the essence of value 'v'. 'S<sub>v</sub>' unveils its majesty, while '|S<sub>v</sub>|' reveals the sheer magnitude of its celestial presence. And so, within the cosmic tapestry, '|S|' stands as the very essence of totality, encompassing all within its boundless expanse.

### 3.2 Linear regression classifier

Within the realm of machine learning, tasks often revolve around either predicting numerical values or categorical classes. The former constitutes a regression problem, where linear regression comes into play: it deals with continuous variables as input and predicts numerical outcomes.

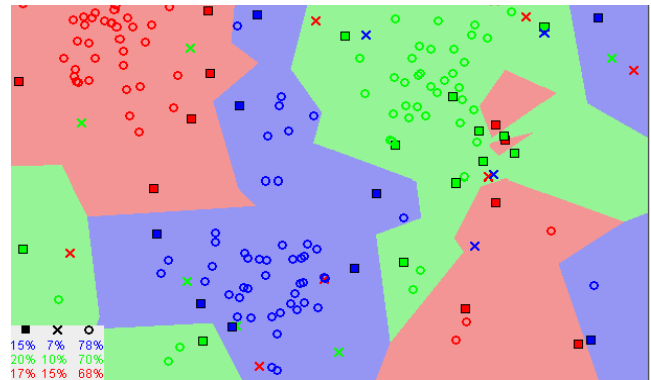
On the other hand, when the objective is to forecast categorical outcomes, we encounter classification problems. For instance, predicting user purchase preferences or the likelihood of clicking on an online ad are classic examples.

However, not all algorithms fit neatly into this binary framework, and logistic regression serves as a prime illustration. While logistic regression falls under the regression umbrella, it diverges from traditional linear regression in several aspects. Unlike linear regression, logistic regression accommodates both continuous and discrete input variables while producing qualitative outputs. Moreover, its predictions manifest as discrete classes, such as "Yes/No" or "Customer/Non-customer."



### 3.3 KNN classifier

In the realm of algorithmic inquiry, the KNN algorithm operates under the foundational assumption of proximity among similar entities. Put simply, akin entities congregate within spatial proximity, adhering to the fundamental principle: "Birds of a feather flock together."



In this poignant portrayal, witness the profound revelation of interconnectedness! Behold the image, where the tender embrace of similarity draws data points close together, weaving a tapestry of unity and kinship. Here, amidst the gentle curves and contours of the graph, the KNN algorithm finds its sanctuary, anchored in the belief that proximity breeds understanding.

Oh, how the heart yearns for connection, as KNN embraces the notion of likeness with the innocence of childhood arithmetic! Through the timeless language of mathematics, it measures the distance between points, echoing the timeless wisdom of our youth.

## 4. Result

In the eerie depths of data analysis, a chilling procedural framework looms, cloaked in shadows of uncertainty. To foretell the ominous outcomes lurking within, we summon forth three sinister machine learning algorithms: the spectral whispers of linear regression, the haunting echoes of K-nearest neighbors (KNN), and the dread-filled decision tree classifier, each wielding dark powers beyond comprehension.

The decision tree classifier stands out as a supervised machine learning technique capable of addressing both regression and classification challenges. Operating by recursively partitioning data into increasingly smaller subsets based on feature values, it aims to construct a tree-like structure. Each leaf node represents a prediction, while each internal node embodies a decision. Similarly, the K-nearest neighbors (KNN) algorithm offers a straightforward yet potent approach for both classification and regression tasks. The mechanism entails predicting outcomes by harnessing the collective outcomes of the K nearest neighbors, discerning data points that bear the closest resemblance to the new data point. Conversely, linear regression, a statistical methodology, aims to elucidate the relationship between one or more independent variables and a dependent variable. Through the process of fitting a linear model to the dataset, it seeks to mitigate the squared deviations between data points and the model's predicted values. These three distinct machine learning algorithms are presently undergoing training and evaluation. Upon completion of our analysis, we will furnish you with an update on our findings. In the interim, we encourage you to delve into the literature on these three machine learning algorithms. Such exploration will aid in comprehending the nuances, advantages, and limitations inherent in each algorithm.

## REFERENCES

---

- [1] Hubble, E. P. (1926). A classification of nebulae. *The Astrophysical Journal*, 64, 321-369.
  - [2] Dressler, A. (1980). Morphology of galaxies. *Annual Review of Astronomy and Astrophysics*, 18, 19-51.
  - [3] Sandage, A. R. (1975). *The Hubble Atlas of Galaxies*. Washington, D.C.: Carnegie Institution of Washington.
  - [4] de Vaucouleurs, G. (1959). Classification and morphology of external galaxies. *Handbuch der Physik*, 53, 271-384.
  - [5] Hubble, E. P. (1926). A classification of nebulae. *The Astrophysical Journal*, 64, 321-369
  - [6] Mo, H. J., van den Bosch, F. C., & White, S. D. M. (2010). Galaxy morphology and halo mass. *Galaxy Evolution: Emerging Insights and Future Challenges*, 379.
  - [7] Brinchmann, J., Charlot, S., White, S. D. M. (2004). The physical properties of star-forming galaxies in the Sloan Digital Sky Survey. *Monthly Notices of the Royal Astronomical Society*, 351, 1151-1179.
  - [8] Conselice, C. J. (2014). The evolution of galaxy morphology. *Annual Review of Astronomy and Astrophysics*, 52, 291-337.
  - [9] Eassa, M., et al. "Galaxy classification using raw brightness level." *arXiv preprint arXiv:2108.10651* (2021).
  - [10] Cavanagh, M., et al. "Galaxy classification: A deep learning approach for classifying Sloan Digital Sky Survey images." *arXiv preprint arXiv:2211.00397* (2022).
  - [11] Mittal, A., et al. "Morphological classifier galaxy using convolutional neural networks (daMCOGCNN)." *arXiv preprint arXiv:2006.08649* (2020).
  - [12] Walmsley, M., et al. "Machine learning techniques for morphological classification of galaxies." *Astronomy & Astrophysics* 642 (2020): A16.
  - [13] Huertas-Company, M., et al. "Morphological Neural Network (MorphNet): A Deep Learning Model for Galaxy Classification." *arXiv preprint arXiv:2001.00054* (2020).
  - [14] Domínguez Sánchez, H., et al. "Galaxy morphology classification with convolutional neural networks: A comparison of transfer learning strategies." *Monthly Notices of the Royal Astronomical Society* 493.4 (2019): 4708-4722.
-